Synthesis

Chapter

5

1. *6.2.1 (2016 SP:22)*

Ethene (CH_2CH_2) can be used to manufacture ethyl ethanoate, $CH_3COOCH_2CH_3$ in three steps, as indicated below:

$$\boxed{CH_2CH_2} \xrightarrow{\text{Step 1}} \boxed{A} \xrightarrow{\text{Step 2}} \boxed{B} \xrightarrow[\text{esterification}]{\text{Step 3}} \boxed{CH_3COOCH_2CH_3}$$

Which one of the following is the correct sequence of Steps 1 and 2?

	Step 1	**Step 2**
(a)	substitution with water	oxidation
(b)	addition of water	oxidation
(c)	oxidation	addition of water
(d)	oxidation	substitution with water

2. *6.2.4 (2016:05)*

Which one of the following substances completes the equation illustrating the process of transesterification to produce biodiesel?

$$\text{Triglyceride} + \boxed{\qquad\qquad} \rightarrow \text{ester (biodiesel)} + \text{glycerol}$$

(a) sodium hydroxide

(b) methanol

(c) sodium methoxide

(d) lipase

3. *6.2.9 (2016:22)*

A soap solution can be used to measure the hardness of a water sample. Four 100.0 mL water samples were tested. The table below shows the results of the tests on the four samples.

Sample	Boiling point (°C) at atmospheric pressure	Volume (mL) of soap solution required to form a permanent lather	Mass (g) of precipitate formed when excess silver nitrate solution is added
A	101.7	17.2	1.2
B	100.3	2.1	0.9
C	100.6	4.2	0.6
D	102.4	9.3	1.4

Which of the four samples contained water with the greatest hardness?

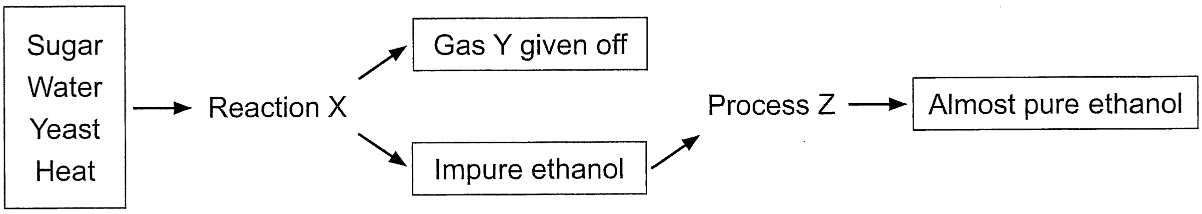
(a) Sample A

(b) Sample B

(c) Sample C

(d) Sample D

Questions *(2016:23)*, *(2016:24)* and *(2016:25)* relate to the flow diagram below showing a process for making ethanol.



4. *6.1 (2016:23)*

Reaction X is called

(a) neutralisation.

(b) fermentation.

(c) condensation.

(d) esterification.

5. *6.1 (2016:24)*

Gas Y is

(a) oxygen.

(b) hydrogen.

(c) carbon dioxide.

(d) carbon monoxide.

6. *6.1 (2016:25)*

Process Z is

(a) fractional distillation.

(b) condensation.

(c) filtration.

(d) precipitation.

7. *6.1 (2017:11)*

The purpose of 'green chemistry' is to

(a) utilise renewable energy sources (such as wind, solar or wave) at all times, even if they are more costly.

(b) design chemical products and processes that maximise profits and, if economical to do so, reduce harm to the environment.

(c) design chemical products and processes that work most efficiently.

(d) design safer chemical products and processes that reduce or eliminate the use and generation of hazardous substances.

8. *6.2.1 (2017:13)*

Consider the following reaction sequence.
Step 1: $H_2O_2(aq) + I^-(aq) \rightarrow H_2O(\ell) + IO^-(aq)$

Step 2: $H_2O_2(aq) + IO^-(aq) \rightarrow H_2O(\ell) + O_2(g) + I^-(aq)$

At the same temperature, Step 1 consumes $H_2O_2(aq)$ at a rate of 4.55×10^{22} molecules per second and Step 2 consumes $H_2O_2(aq)$ at a rate of 3.67×10^{24} molecules per second.

The only source of $IO^-(aq)$ for Step 2 comes from the reaction in Step 1.
Which one of the following statements identifies, with justification, the rate-determining step?

(a) Step 1, because it is the first step in the sequence.

(b) Step 2, because it is the last step in the sequence.

(c) Step 1, because it is the slower step in the sequence.

(d) Step 2, because it is the faster step in the sequence.

9. *6.2.8 (2017:21)*

The structure of detergent could be represented as R-X where R is dodecylbenzene and X is a sulfonate. X is represented by which one of the following?

(a) SO_4^{2-}

(b) SO_3^{2-}

(c) SO_3^-

(d) SO_2^-

Questions *(2018:19)* and *(2018:20)* refer to two different methods of synthesising ethanol.

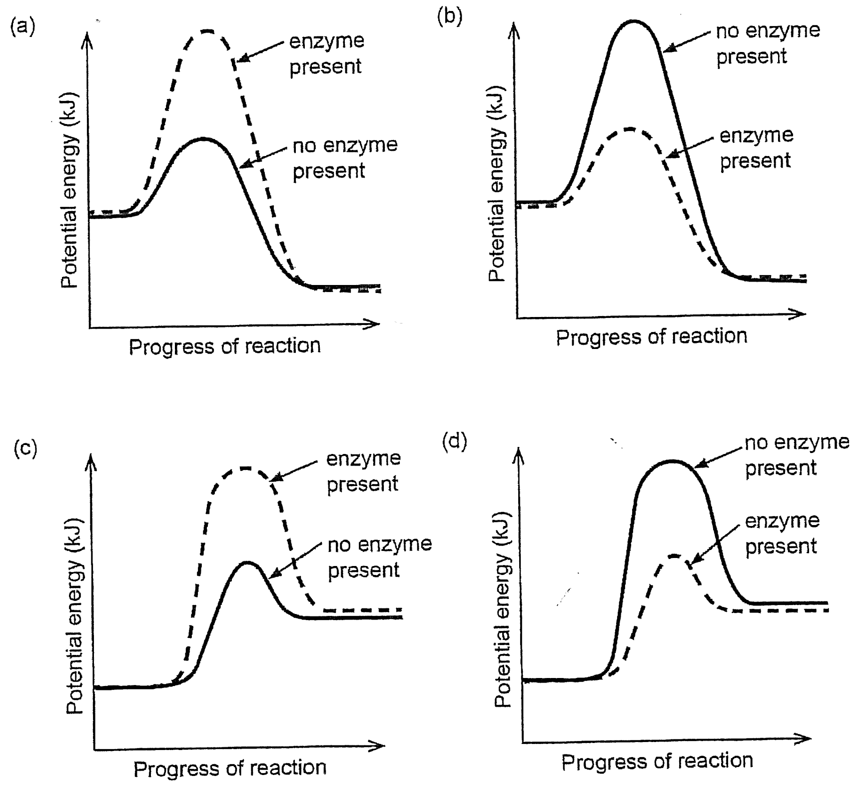
10. *6.2.5 (2018:19)*

Ethanol can be synthesised by combining sugar, a suitable enzyme and water. The equation for this is shown below.

$$C_6H_{12}O_6(aq) \rightleftharpoons 2\ CH_3CH_2OH(aq) + 2\ CO_2(g) + \text{energy}$$

This process is known as 'fermentation'.

Which of these energy profile diagrams shows the effect of an enzyme on this reaction? All other conditions are constant.



11. *6.2.5 (2018:20)*

Ethanol can also be produced by reacting ethene with steam. The equation for this reaction, which is known as hydration, is shown below.

$$CH_2 = CH_2(g) + H_2O(g) \rightarrow CH_3CH_2OH(g) \qquad \Delta H = -45 \text{ kJ mol}^{-1}$$

Which statement about this reaction is **incorrect**?

(a) The hydration reaction requires higher pressures than fermentation to achieve an economically-viable rate.

(b) The hydration of ethene is an addition reaction.

(c) Enzymes decrease the activation energies of both the hydration and fermentation reactions.

(d) The hydration reaction requires higher temperatures than fermentation to achieve an economically-viable rate.

Questions *(2018:21)*, *(2018:22)* and *(2018:23)* relate to the following information.

Ammonia, NH_3, is an industrially-important chemical. It is produced on an industrial scale by the Haber process. The reaction for the Haber process is shown below.

$$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g) + 92 \text{ kJ mol}^{-1}$$

12. *6.2.4 (2018:21)*

What is the immediate effect of increasing the temperature on the rates of the forward and reverse reactions in the Haber process?

(a) The rates of the forward and reverse reactions increase equally.

(b) The rates of both reactions increase while the rate of the reverse reaction increases more than the rate of the forward reaction.

(c) The rates of both reactions increase while the rate of the forward reaction increases more than the rate of the reverse reaction.

(d) The rate of the forward reaction remains unchanged while the rate of the reverse reaction increases.

13. *6.2.4 (2018:22)*

What combination of temperature and pressure should be used to maximise the yield of ammonia, NH_3?

	Temperature for maximum NH_3 yield	Pressure for maximum NH_3 yield
(a)	high temperature	low pressure
(b)	high temperature	high pressure
(c)	low temperature	low pressure
(d)	low temperature	high pressure

14. *6.2.4 (2018:23)*

The Contact process, which is used to produce sulfuric acid, is another industrially-important process. The process contains several steps, one of which is the production of sulfur trioxide, SO_3, as shown below.

$$2\ SO_2(g) + O_2(g) \rightleftharpoons 2\ SO_3(g) \qquad \Delta H = -196\ kJ\ mol^{-1}$$

Which statement regarding both the Haber and Contact processes is correct?

(a) Both are exothermic and both need a suitable catalyst to occur at a satisfactory rate.

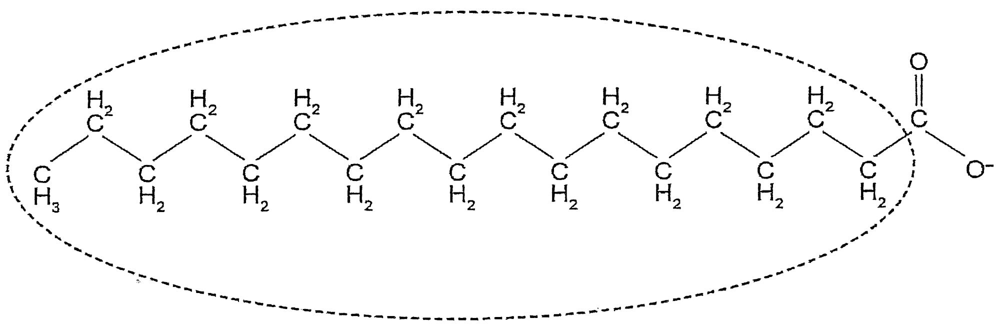
(b) Both are endothermic and both need a suitable catalyst to occur at a satisfactory rate.

(c) Both need a suitable catalyst so that the yield of their respective products is maximised.

(d) Both can achieve high rates and high yields without the need for a catalyst.

15. *6.2.8 (2018:24)*

Soap and water can be used to remove oil and grease from human skin. The following diagram represents an anion of soap.



Which of the following correctly describes the orientation of the circled section and the attractive force it forms?

	Orientation	Attractive force
(a)	toward oil and grease	dispersion forces
(b)	toward oil and grease	dipole-dipole forces
(c)	toward water	hydrogen bonds
(d)	toward water	ion-dipole forces

16. *6.2.4 (2019:11)*

Which one of the following statements about catalysis in the production of biodiesel is correct?

(a) Base catalysis generally has a higher reaction rate but, unlike lipase catalysis, can cause saponification, which decreases the biodiesel yield.

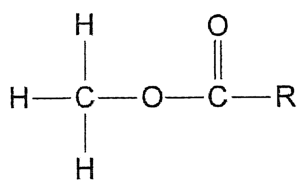
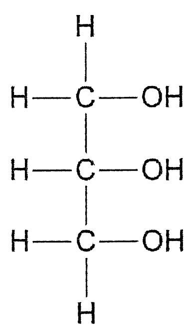
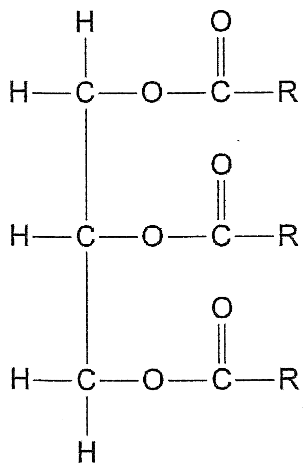
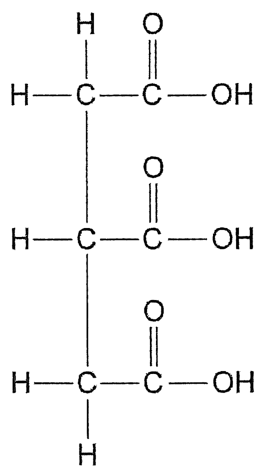
(b) The sodium hydroxide and potassium hydroxide used in base catalysis are readily available and relatively cheap, but lipase catalysis produces more toxic waste water.

(c) Base catalysis involves only one step, while lipase catalysis involves many steps in its synthesis sequence, which in turn adds to the cost of the process.

(d) Base catalysis typically has a lower rate and yield of biodiesel but lipase catalysis is sensitive to alcohols, such as methanol, and has higher energy costs.

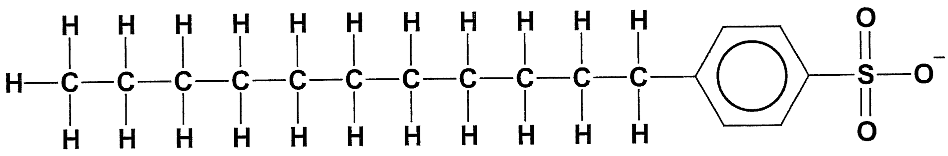
17. *6.2.4 (2019:13)*

One method of producing biodiesel is by a transesterification reaction where triglycerides are converted into simpler methyl esters (the biodiesel) of the fatty acids. Which one of the following is a reactant of this transesterification reaction?

(a)		(b)	
(c)		(d)	

18. *6.2.8 (2020:09)*

Consider the molecule shown below.



This molecule shows

(a) an anionic detergent which contains a sulfonate group.

(b) a monomer that can be used to synthesise a condensation polymer.

(c) a carboxylic acid which can be used to synthesise a soap.

(d) an aromatic hydrocarbon which has donated an electron.

19. *6.2.9 (2020:22)*

When cleaning greasy/dirty objects in hard water, it is **best** to use

(a) a soap, because it forms a precipitate with the ions causing water hardness, thereby removing these ions from solution.

(b) a detergent, because it does not react with the ions causing water hardness.

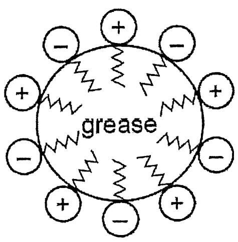
(c) a detergent, because it forms a precipitate with the ions causing water hardness, thereby removing these ions from solution.

(d) a soap, because it does not react with the ions causing water hardness.

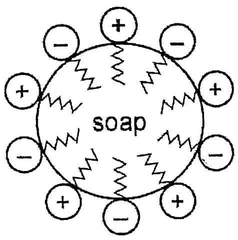
20. *6.2.8 (2020:23)*

Which of the following diagrams represents the micelle that forms in water when soap is used to remove grease from dirty dishes?

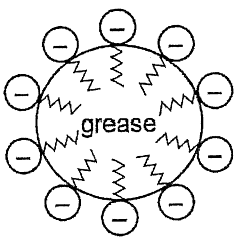
(a)



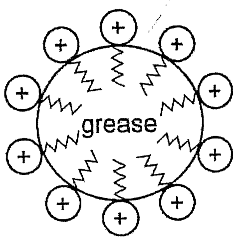
(b)



(c)



(d)



determine oxidation outcomes (aldehyde, ketone). The presence of double bonds similarly changes chemical and physical properties. The answer is (d).

33.(2020:11) A peptide (or amide) is a condensation product of an amine and a carboxylic acid that can be used to form a polymer. The constituent molecules should have suitable functional groups either $-NH_2$ or $-COOH$ must be present. The answer is (b).

34.(2020:13) All of these alcohols can demonstrate hydrogen bonding with each other. Pentan–1–ol molecules can pack closest together. Therefore, they will be the hardest to separate requiring more energy that is available at a high temperature. The answer is (a).

35.(2020:14) The structure (shape) of a protein determines its function like a key in a lock. The answer is (d).

36.(2020:21) This polymer is an example of an addition polymer. The repeating unit is $CH_2{=}CH{-}CN$. The answer is (c).

Chapter 5: Synthesis

1.(2016 SP:22-b) To form an ester an alcohol and a carboxylic acid are required. The ethene can form ethanol by an addition reaction in the presence of water, and the ethanol can be oxidised to form ethanoic acid. These two substances can react to form ethyl ethanoate. The answer is 'b'.

2.(2016:05-b) This is an equation mentioned in syllabus and should be known. NaOH produces soap while methanol produces the transesterification reaction making biodiesel. The answer is 'b'.

3.(2016:22-a) Hardness is a measure of calcium ion concentration. This is measured by volume of soap reacted. Other salts (chlorides etc.) will respond to silver ions or raise the boiling point but they have nothing to do with hardness. Only soap solution is indicative of hardness. The answer is 'a'.

4.(2016:23-b) Reaction X is a REDOX reaction with catalyst but this is not one of the answers offered. The best answer is 'b'.

5.(2016:24-c) This is a known reaction from the syllabus. The answer is 'c'.

6.(2016:25-a) This is a known reaction from the syllabus. The answer is 'a'.

7.(2017:11-d) There are 12 principles of green chemistry you should have read. The minimising toxic waste and by-products is essential. The answer is 'd'.

8.(2017:13-c) Rate determining steps have not been examined previously in the WACE. The idea is that in a sequence of reactions the overall reaction rate is dependent on the slowest step. The answer is 'c'.

9.(2017:21-c) The syllabus states 'the structure of the anionic detergents derived from dodecylbenzene contains a non-polar hydrocarbon chain and a sulfonate group'. You should know what a sulfonate group is. The answer is 'c'.

10.(2018:19-b) An enzyme lowers the activation energy so (a) and (c) are wrong. The reaction is exothermic so the answer is 'b'.

11.(2018:20-c) Answers (a), (b) and (d) are all **correct**. The hydration reaction does not use enzymes. The answer is 'c'.

12.(2018:21-b) The reverse reaction has a large Ea due to the fact the reaction is exothermic. Thus, the reverse reaction benefits more from the use of a catalyst than the forward reaction which has a relatively small Ea. The answer is 'b'.

13.(2018:22-d) The question asks only about yield. The reaction is exothermic thus removing heat will favour the forward reaction and the mol ratio indicates high pressure will also favour the forward reaction. The answer is 'd'.

14.(2018:23-a) Both reactions are exothermic and the Haber process uses a catalyst mostly made of iron while the Contact process uses vanadium pentoxide as a catalyst but remember catalysts do not alter yield – just rate. The answer is 'a'.

15.(2018:24-a) The long hydrocarbon chain demonstrates substantial dispersion forces with oil (another molecule with high dispersion forces). The long carbon chain has no lone pairs of electrons and so cannot engage in hydrogen bonding and since it has no ions or dipoles the answer is 'a'.

16.(2019:11-a) This question required detailed understanding of the two biodiesel processes and soap making along with some green chemistry. The answer is 'a'.

17.(2019:13-c) This question again required knowledge of the biodiesel process and required recognition of a triglyceride – (a) and (b) are products of the reaction. The answer is 'c'.

18.(2020:09) This is the standard structure of an alkyl (the straight chain) benzyl (the benzene ring) sulfonate (the negatively charged SO_3^-) – it is a detergent. The answer is (a).

19.(2020:22) Soaps form precipitates with calcium and magnesium ions dissolved in water. To avoid this problem, we use detergents because they do not form precipitates. The answer is (b).

20.(2020:23) In soap, for example, sodium stearate, the stearate ion is negatively charged. The long carbon chain is attracted to the grease by dispersion forces to form a micelle leaving the ion $-COO^-$ exposed to interact with water molecules. The answer is (c).

Chapter 6: Science Enquiry Skills

1.(2016 SP:11-b) Systematic error is caused by faulty equipment or faulty methodology and cannot be reduced by repeating an experiment. A pipette offers more sig figs and will lower random error. Answer is 'b'.

2.(2016:06-d) A systematic error can be identified by data that is precise (many values close together) but wrong. The answer is 'd'.

3.(2017:22-d) The concentration of $HC\ell$ is varied and the effect on the other variables are measured. This makes it the independent variable. The answer is 'd'.

4.(2017:23-c) The data involved numbers (quantitative) and was collected by the experimenter (thus it was primary data). The answer is 'c'.

5.(2017:24-a) The data was consistently wrong which means it was not random error. It was systematic caused by a wrong technique or poor equipment. The answer is 'a'.

6.(2017:25-d) The technique would mean there were differing methods used and the data would be less reliable. The answer is 'd'.